Biology

Topic: Plant Kingdom

Introduction

1. Artificial system of classification
1) Includes features like anatomy, morphology and photochemistry
2) Was given by George Bentham and Joseph Dalton Hooker
3) Gave equal weightage to vegetative and floral characters
4) Brought together closely related species

2. System of classification proposed by Bentham and Hooker
1) Is based on natural affinities among the organisms
2) Believes that organisms belonging to same taxa have a common ancestor
3) Considered embryology, phytochemistry and anatomy of plants as criteria
4) Both (1) and (3)

3. The classification of organisms on the basis of DNA sequencing is known as
1) Cytotaxonomy
2) Chemotaxonomy
3) Phenetics
4) Adansonian taxonomy

4. Which of the following feature was not used by Bentham and Hooker for classification of plants?
1) External features
2) Anatomy
3) Phytochemistry
4) Phylogeny

5. The system of classification which has separated closely related species as they are based on few characteristics is
1) Phylogenetic system
2) Artificial system
3) Natural system
4) Both (1) and (3)

6. Match the following columns and choose the correct option

Column I		Column II	
a	Numerical taxonomy	1	Cytological information
b	Cytotaxonomy	2	Chemical constituents of the plant
c	Chemotaxonomy	3	Easily carried out using computers

1) a-3, b-2, c-1
2) a-3, b-1, c-2
3) a-2, b-1, c-3
4) a-1, b-3, c-2

7. Which of the following features is common in artificial, natural and phylogenetic system of classification?

1) Anatomy
2) Phylogeny
3) Morphology
4) Phytochemistry

8. Phylogenetic classification system assumes that

1) Organisms belonging to different taxa have a common ancestor
2) Organisms belonging to the same taxa have different ancestor
3) Organisms belonging to the same taxa have a common ancestor
4) All of the above

Algae

9. Read the following statements w.r.t. algae and select the correct option.

I. They are simple, thalloid and largely aquatic.
II. They occur in variety of habitats including moist stone, soil or wood.
III. The form and size of algae are highly constant.

1) Only (I) is correct
2) Only (I) and (II) are correct
3) Only (I) and (III) are correct
4) All (I), (II) and (III) are correct

10. Which of the following features are correct for algae?

I. Unicellular and non-jacketed sex organs.
II. Well developed and differentiated sporophyte.
III. Presence of embryo.
IV. Presence of motile male gametes.

1) I and II
2) I and III
3) I and IV
4) III and IV

11. The main criterion used in classification of algae is

I. Flagellation
II. Type of pigmentation
III. Chemistry of cell wall

1) II only
2) II and III only
3) I and II only
4) I, II and III

12. How many of the following features are associated with members of red algae?

Laminarin and mannitol, Non-motile spores, Oogamous reproduction, Phycocolloids in cell wall, Complex post

fertilization development

1) Three
2) Four
3) Two
4) Five

13. The algae which are mostly found in marine habitats with abundance in the warmer region

A. Possess chlorophyll a and d
B. Show isogamous sexual reproduction
C. Do not show post-fertilization development
D. Are characterised by absence of motile spores and gametes

Choose the correct option.

1) Only (A) and (C) are correct
2) Only (C) is incorrect
3) Only (A) and (D) are correct
4) (A), (B), (C) and (D) are correct

14. Select incorrectly matched pair

1) Food source – Laminaria
2) Agar- Gelidium and Gracilaria
3) Carrageen - Brown algae
4) Space travelers food supplement - Chlorella

15. How many of the following structures in the algae are motile?

Zoospore, Chlamydospores, Akinetes, Gametes ofChlamydomonas, Gametes of Spirogyra, Gametes of Fucus, Gametes of Porphyra, Aplanospores

1) Three
2) Four
3) Five
4) Two

16. Read the following statements and select the correct option:

Statement I: The red thalli of most of the red algae are multicellular and some of them have complex body organisation.

Statement II: Food stored in red algae is complex carbohydrates which may be in the form of laminarin or mannitol.

Choose the correct option:

1) Only statement (I) is correct
2) Only statement (II) is correct
3) Both statements (I) and (II) are correct
4) Both statements (I) and (II) are incorrect

17. Consider the following statements.

A. They vary in colour from olive green to various shades of brown depending upon the amount of xanthóphyll present.
B. Present at a great depths in the oceans where relatively little light penetrates.

C. Their sexual reproduction shows considerable variation in the type and formation of sex cells.

These statements are correct for which class of algae?

	A	B	C
1)	Phaeophyceae	Chlorophyceae	Rhodophyceae
2)	Phaeophyceae	Rhodophyceae	Chlorophyceae
3)	Chlorophyceae	Rhodophyceae	Phaeophyceae
4)	Rhodophyceae	Chlorophyceae	Phaeophyceae

Bryophytes

18. Read the following statements w.r.t. bryophytes and select the correct option:

I. Haploid phase in the life cycle is more differentiated than that of algae.

II. Zygote divides by meiosis immediately.

III. The sporophyte is not free living and independent.

1) Only statement I is correct
2) Statements II and III are correct
3) Statements I and III are correct
4) All statements are correct

19. Given below are two statements: one is labelled Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Bryophytes are also called amphibians of the plant kingdom.

Reason (R): These plants can live in soil but are dependent on water for sexual reproduction.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

20. Members of liverworts

A. Grow usually in moist and shady habitats.

B. Have free living sporophyte.

C. Reproduce asexually by fragmentation of protonema.

D. Reproduce asexually by gemmae.

Choose the correct option to complete the sentence

1) Only (D) is correct
2) (A) and (D) are correct
3) (A), (C) and (D) are correct
4) (A), (B), (C) and (D) are correct

21. Rhizoids of mosses are

1) Multicellular and branched
2) Multicellular and unbranched

3) Unicellular and unbranched
4) Not found in mosses

22. Gemma is
1) Multicellular sexual buds
2) Green, multicellular asexual buds
3) Asexual unicellular buds of mosses
4) Present in large receptacles called gemma cups located on the leaf appex

23. Fill in the blanks correctly w.r.t. the life cycle of a moss:

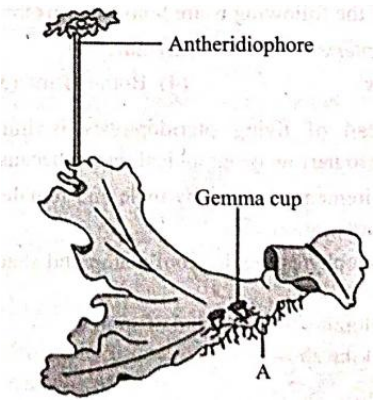
A moss sporophyte $\longrightarrow$ A $\longrightarrow$ Spore mother cell $\xrightarrow{B}$ Spore $\longrightarrow$ C $\longrightarrow$ Leafy gametophyte

	A	B	C
1)	Capsule	Meiosis	Protonema
2)	Sporangia	Meiosis	Protonema
3)	Capsule	Meiosis	Prothallus
4)	Sporangia	Meiosis	Gametophyte

24. Select the incorrect statements w.r.t. bryophytes:
1) They produce only one type of spores.
2) They play an important role in biological succession on rocks.
3) They have great economical importance.
4) Sporophyte is not true living.

25. Juvenile, filamentous stage of moss
1) Is called protonema
2) Develops from germination of spore
3) Reproduces by fragmentation
4) All of these

26. Identify the structure labelled as 'A'.



Choose the correct option.
1) Rhizoids
2) Mycorrhiza
3) Adventitious roots
4) Root hairs

Pteridophytes

27. In pteridophytes, the sporophytic plant body

A. Is diploid with most primitive vascular tissues.

B. Is differentiated into root, stem and leaf and bears sporangia.

Select the correct option to complete the sentence.

1) Only (A) is correct
2) Only (B) is correct
3) Both (A) and (B) are correct
4) Both (A) and (B) are incorrect

28. Pteridophytes do not form

1) Cones
2) Prothallus
3) Spores
4) Seeds

29. Which of the following is not related to importance of pteridophytes?

1) Used as medicine
2) Act as soil binder
3) Used in ornamental purposes
4) Forms peat

30. Pteridophytes have dominant sporophytic phase with

1) Primarily aquatic habitat
2) Advanced type of vascular tissue
3) Sporangia bearing leaf like appendages called sporophyll
4) Undifferentiated plant body

31. Which of the following is/are homosporous in nature?

1) Dryopteris
2) Salvinia
3) Pteris
4) Both (1) and (3)

32. The spread of living pteridophytes is limited and restricted to narrow geographical region because

1) Requirement of water by male and female gametes for fertilization
2) Gametophytes require cool, damp and shady places to grow
3) Development of two kind of spores
4) All of the above

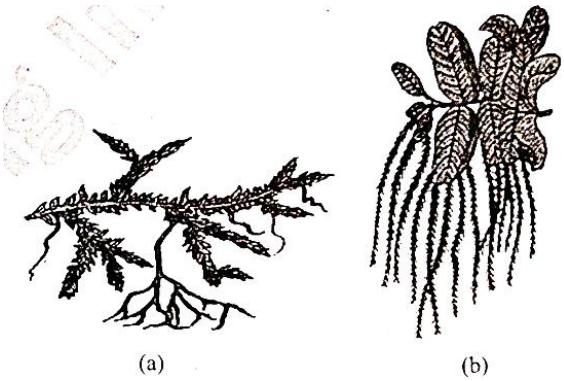
33. Which of the following feature/event in pteridophytes is a precursor to the seed habit considered as an important step in evolution?

1) Development of cone
2) Vascular tissue
3) Internal fertilization
4) Heterospory

34. Select the incorrect match w.r.t. pteridophytes:

1) Small leaves – Selaginella
2) Large leaves – Ferns
3) Compactly arranged sporophylls – Equisetum
4) Vegetatively propagated by leaftip - Salvinia

35. Which of the given features are common for the following plants?



1) Homosporous
2) Absence of vascular tissue
3) Development of zygote into embryo within female gametophyte
4) Aquatic habitat

36. In some pteridophytes sporophylls may form distinct compact structures called strobili or cones like in

1) Sellaginella
2) Equisetum
3) Funaria
4) Both (1) and (2)

Gymnosperms

37. Read the following statements w.r.t. gymnosperms and state true (T) or false (F):

A. Include medium size trees or tall trees and shrubs of perennial habit.
B. Ovules remain exposed, both before and after fertilization.
C. All the members possess tap root system in association with fungal filament.
D. It is the smallest group of plant kingdom.

	A	**B**	**C**	**D**
1)	T	T	F	F
2)	T	T	F	T
3)	T	F	T	F
4)	F	F	T	T

38. Select incorrect statement w.r.t. naked seeded plants.

1) Reduced gametophytes having independent free-living existence.
2) They are heterosporous
3) Dominant phase is sporophyte.
4) They exhibit diplontic life cycle pattern.

39. Read the following statements carefully w.r.t. gymnosperms:

A. Thick cuticle and sunken stomata present in the leaves of gymnosperms help to reduce water loss.
B. All gymnosperms are heterosporous.
C. Sequoia is one of the tallest gymnosperms.
D. In Pinus, the male and female gametophytes have free-living existence.

How many of the above given statements are correct?

1) Two
2) Three
3) Four
4) One

40. Which of the following structure is not present in Cycas?

1) Unbranched stem
2) Coralloid roots
3) Pinnate leaves
4) Female cone

41. Read the following statement w.r.t. leaves of gymnosperm:

A. In gymnosperms the leaves may be simple or compound.
B. Sunken stomata and thick cuticle are found to reduce the rate of transpiration.
C. The leaves are well adapted to withstand extremes of temperature, humidity and wind.
D. In conifers, the needle-like leaves reduce the surface area

Select the correct statements:

1) C and D
2) A, B, Cand D
3) A and B
4) A, B and D

42. Adaptation in conifers to reduce water loss is/are presence of

1) Thick cuticle
2) Sunken stomata
3) Needle-like leaves
4) All of these

43. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Micro and megasporophylls are borne on different plants in Cycas.

Reason (R): Cycas is a dioecious gymnosperm.

1) Both A and R are true and the R is the correct explanation of the A
2) Both A and R are true, but the R is not the correct explanation of the A
3) A is true but R is false
4) Both A and R are false

44. Consider the following statements and select the correct option w.r.t. Cycas:

A. Unbranched stem
B. Dioecious plant

C. Leaves are simple, needle like
D. Presence of coralloid roots
E. Pollen grains are carried by air and insects
1) A, B, D
2) A, C, D
3) A, B, D, E
4) A, B, C, D, E

45. Ploidy levels of endosperm and embryo in gymnosperms are, respectively
1) n, 2n
2) 2n, n
3) 3n, 2n
4) n, n

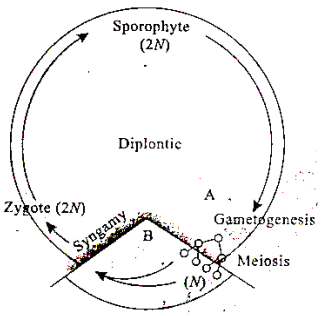
46. Gymnosperm differs from angiosperm in
1) Absence of fruits
2) Presence of diplontic life cycle pattern
3) State of meiosis
4) Presence of integumented megasporangium

47. Consider the following statements about the sporophytic generation:
A. The generation that produces the spores.
B. The generation that possesses vascular tissue.
C. The diploid generation.
Correct statements are
1) Only (A)
2) Only (C)
3) Only (A) and (C)
4) (A), (B) and (C)

48. Which of the following is not having a haplontic life cycle?
1) Fucus
2) Chlamydomonas
3) Ulothrix
4) Spirogyra

49. Choose the odd one w.r.t. life cycle pattern.
1) Chlamydomonas
2) Volvox
3) Fucus
4) Spirogyra

50. All statements are true for the given life cycle pattern, except



1) All spermatophytes follow this life cycle pattern.
2) Gametic meiosis.
3) The gametophytic phase is represented by single to few celled haploid gametophyte.
4) The main plant body is photosynthetic independent sporophyte.

51. Which of the following features correctly belong to the concerned plant group?

A. Presence of seed habit.

B. Sporophyte is not independent, but it is attached to photosynthetic gametophyte.

C. Endosperm is haploid and develops before fertilization.

	A	B	C
1)	Pteridophyte	Bryophyte	Angiosperm
2)	Pteridophyte	Bryophyte	Gymnosperm
3)	Pteridophyte	Gymnosperm	Angiosperm
4)	Bryophyte	Gymnosperm	Angiosperm

52. Given below are two statements

Statement I: Following fertilisation, the zygote also divides by meiosis to produce a diploid sporophytic plant body.

Statement II: Spores divide by meiosis to form a haploid plant body.

In the light of the above statements, choose the correct answer from the options given below:

1) Statement I is correct, but statement II is incorrect.
2) Statement I is incorrect, but statement II is correct.
3) Both statement I and statement II are correct.
4) Both statement I and statement II are incorrect.

53. Sporophytic generation represented by only single cell, i.e. zygote is seen in

1) Mosses
2) Algae
3) Ferns
4) Conifers

54. During the life cycle of any sexually reproducing plant, there is an alternation of generations between

1) Gamete producing haploid sporophyte and spore producing diploid gametophyte
2) Gamete producing diploid gametophyte and spore producing haploid sporophyte

3) Gamete producing diploid sporophyte and spore producing haploid gametophyte
4) Gamete producing haploid gametophyte and spore producing diploid sporophyte

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
3	4	2	4	2	2	3	3	2	3	4	2	3	3	1
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
1	2	3	4	2	1	2	1	3	4	1	3	4	4	3
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
4	2	4	4	3	4	2	1	2	4	2	4	1	1	1
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**						
1	4	1	3	2	2	4	2	4						